EGB320 Final Report: Mobility Subsystem

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1 Problem Description

The problem set forward by the client was that shelves needed restocking, and it was believed that a robotic system would provide a more cost-effective method for shelf restocking. To evaluate the suitability of a robotic warehousing system a Technology Readiness Level 3(TRL3) prototype. This prototype was designed to work in a scaled down mock warehouse with shelves, items and obstacles that would simulate the major factors around navigating and moving stock around a grocery store.

For this prototype some adjustments were made to allow for more streamlined development. The first major adjustment was to have each different element in the store be represented by an object of a different color and shape as shown below:

Element	Representation
Obstacle	Green Colouring
Item	Orange Colouring
Aisle and Picking Station Marker	Black Colouring
Shelves	Blue Colouring
Picking Station	Yellow Colouring

These adjustments were made to allow for a more simplified vision system which in turn would allow for rapid development. Further prototypes will build on this model to allow for more complex vision processing. Along with this the shelf heights were the same for all aisles. Allowing for simplified item placement and navigation code as no matter the bay the heights will always remain the same. The final simplification is the distance between the picking stations, this distance allows for a lower level accuracy being required to pick up an item.

With these simplifications in mind 4 separate subsystems were identified that would be designed to integrate to a final product. The intention behind this was to further simplify the system through having each system capable of operating separately with methods of communication to allow for integration. This allowed for the 4 subsystems to be designed in parallel and setbacks in one element of the robot would not necessarily set back the overall progress of the robot.

Following this the requirements for each subsystem were identified, for mobility these requirements were as follows:

- Turn to a bearing on a single point in a given direction
- Turn at a set speed in a given direction
- Move forward and backward at a set speed
- Move forward and backward a set distance
- Have a chassis that is capable of carrying all other subsystems
- Have a power management system that can power all other subsystems

With these requirements identified background research was conducted to determine the most suitable solutions.

2 Background Research

Research was conducted into various methods for robotic mobility used in similar applications in industries. Four major methods of mobility were identified:

- Wheels
- Tracks
- Legs
- Mecanum Wheels

An investigation into the pros and cons of each system was conducted to allow for a clear evaluation of each option's suitability. The results of this investigation were compiled into the table seen below.

Feature	Wheels	Tracks	Legs	Mecanum Wheels
Can spin in place	✖	✓	✓	✓
Reliably stable without extra sensors	✓	✓	✖	✓
Requires 2 motors or less	✓	✓	✖	✖
Can easily have speed measured in all directions	✓	✓	✖	✖
Easily programmable	✓	✓	✖	✖

As shown in the table above it was decided that tracks would be investigated as the primary option for mobility. This was due to its ability to navigate varying terrain with small obstacles, while maintaining a simple drive system to ensure a reliable outcome. The Treads also have a high amount of surface friction due to the large surface area in contact with the ground. However, there were concerns about the reliability of tracks as consulting an expert on their usage in the field found that they can be prone to slipping off the retainer system over time. This concern was minimal as the environment that the tracks were being deployed in varied greatly from the environment of a warehouse.

Once the research into the mobility system was done it was crucial to determine a method in which the motors will be controlled. Initially it was thought that the Raspberry Pi utilized for the vision and navigation subsystems would also be used to control the motors. This plan was found to have issues one research found that the Raspberry Pi GPIO pins were only capable of 50mA total for the entire pin header. This was a major concern as the motors can easily draw over 5 times this amount under a standard load.

Further research was conducted and found that the majority of robots that involved a mobility subsystem made use of one or multiple motor controllers. These motor controllers take input signals from a central control system and take power directly from the power management system to drive the motors. This bypasses the 50mA current limit of the GPIO pins and allows for stronger motors to be used.

Once this had been decided the next significant piece of research was to determine a power management system that would be capable of supplying the robot and all of its peripherals with power. As the robot's main power supply was an 8.4V Lithium-ion battery it's voltage would need to be regulated down to the required 5V input for the Raspberry Pi. Multiple options were considered for this regulator component, the first being a simple 5V linear regulator due to its simplicity for implementation. This idea was moved past due to concerns that the with the efficiency of the regulator. Following this research was conducted into the feasibility of a switching regulator. Research found that switching regulators are significantly more efficient in their power regulation, however they add high levels of noise to the circuit which can affect signals in the system. This however wasn't deemed to be a major issue as there were no components that are highly susceptible to noise. With research conducted into all elements required for the mobility system it was then time to begin the actual design and component selection for the robot.

3 Design

The design of the subsystem had multiple individual sections that had to be developed sequentially. These sections included, track selection, motor selection, motor controller selection, power management system design, motor controller selection and motor controller programming. Each section had their own challenges and needs that had to be evaluated and designed around. Each section and its design considerations are detailed below.

3.1 Track Selection

The process of selecting tracks was relatively straightforward as not many of the accessible suppliers that stocked tracks were accessible within a time period that would be suitable for the project timeline. This was a major deciding factor, as without a functional mobility system the testing of navigation on the actual robot could not commence. With this in mind local Australian based suppliers became the preferable option. With this limiting factor it was found that Robot Gear Australia had the widest range of options. These options mainly consisted of Pololu Track sets which came in different size sprockets and track belts.

It was decided that the 30 tooth track set was the most suitable, this was mainly due to the sprocket shaft spacing of 85mm. This combined with the sprocket radius of 35mm gives a total track length of 120mm, 60% of the maximum length of 200mm. This length provides stability from rotational forces caused by the robot driving forward and reverse with a mass held at a height. With a decision made on the track set to be used the next in the design process was motor selection.

3.2 Motor Selection

With the tracks selected some Pololu N20 motors were trialed to drive the system. The main reason behind this was that they were already owned and compatible with the track set. These motors were 50:1 N20 DC motors with magnetic encoders and a max output torque of 400gcm. These trial motors were fitted to a baseplate and powered by an exiting motor driver hat designed to integrate with the Raspberry Pi. Testing found that while the motors had sufficient power output to drive the Raspberry Pi and the mounting baseplate they were unable to move the robot up the necessary inclines with the total predicted system weight. As such output torque in a small format became the priority for motor selection.

With output torque becoming the priority for motor selection it became necessary to calculate the minimum torque required to achieve the required movements. The torque required was calculated using the formula:

$$T_m = \mu_s \times m \times g \times radius$$

The torque was calculated for the system using an estimated robot mass of 900 grams with a coefficient of friction of 0.45 for carpet. With an applied factor of safety of 1.2 the estimated torque required was found to be 850gcm. With this value known research was conducted to find an N20 motor that would be capable of producing the required levels.